

Software evaluation and selection

The project required a reproducible route from Turbomole pointval grid output to publication-quality images of the molecular electrostatic potential (ESP) mapped onto an electron-density isosurface. The visualization program is only one component of that route, but it is the one that had to be chosen first, because the format of everything upstream depends on it.

1.1 Requirements

Five requirements follow from the task. The first three are hard: a program failing any of them cannot produce the required figures at all.

#	Requirement	Why it is required
R1	Reads Gaussian cube	The converter writes cube files. A program that cannot read them is out of scope before anything else is considered.
R2	Maps one volume onto an isosurface of another	The surface is the $\rho = 0.001$ a.u. density isosurface; the colour is the ESP on it. Two separate volumetric datasets, one shaping the surface and one colouring it.
R3	Explicit control of the colour scale	Comparing molecules requires that a given colour means the same potential in every figure. An auto-scaling viewer with no numeric handle cannot guarantee that.
R4	Scriptable, non-interactive	Reproducibility. Figure settings must live in a file, not in a sequence of clicks, and the whole set must be regenerable with one command.
R5	Free, no licence server	The workflow has to be executable by the next person without a purchase or a licence negotiation.

Table 1: Requirements used to assess the candidates.

R4 does the actual filtering. Every program in Table 2 can, with enough clicking, produce one correct picture. Only some can guarantee that the twentieth picture was made with the same settings as the first.

1.2 Candidates

Five general-purpose molecular viewers that read volumetric data were assessed. The comparison is a desk study compiled from vendor documentation and the primary literature.

Program	Licence	Cube	ESP on isosurface	Scripting / headless	Ray tracing
PyMOL	BSD-like, free; the Schrödinger build is commercial	yes	yes isosurface from one map, ramp_new over a second, attached with set surface_color	Python API and .pml;	built in (ray)
VMD 1.9.4 / 2.0	free of charge for academic and internal use only	yes	Yes: both cubes into one molecule with mol addfile, Isosurface representation coloured by Volume	Tcl; vmd -dispdev text -e script.tcl	Tachyon, TachyonInternal, TachyonL-OptiX
UCSF ChimeraX 1.11	free for academic use	yes	Yes: color sample on any surface from a second map	command files (.cxc) and Python; --nogui, --offscreen (Linux)	none documented
Avogadro 2 (2.0.0)	BSD 3-clause, free	yes	Partly: the documented ESP colouring comes from an internal charge model, not from a second cube	Python command scripts, invoked from the GUI menu	via POV-Ray export only
GaussView 6.1.1	Commercial academic single computer 750\$	yes	Yes: New Mapped Surface, density surface coloured by an ESP cube	none documented for the visualization side	none documented

Table 2: Capability comparison. Compiled from the documentation cited in ProjectElaboration.pdf; not from hands-on testing except for PyMOL and VMD. Versions and licence terms checked August 2026.

Notes on the individual programs

PyMOL. The colour ramp is an object in its own right, which is what makes R3 easy: the same explicit range can be attached to every molecule of a batch and written into the generated scene file, so the figure and the interactive scene cannot drift apart. `pymol -cq` is a first-class headless mode.

VMD. Technically the closest competitor and the one named in the project brief. It has no ramp object; instead both grids are loaded into the same molecule ID and the isosurface of the first is coloured by the values of the second. Two practical caveats emerged only in use and neither is visible in the documentation: its ray tracer aborts unpredictably on stacked transparency and `TachyonInternal` renders at window size, so image dimensions are capped by the screen.

The three that were not built. ChimeraX meets all five requirements on paper and would have been a defensible third choice; it was left out for scope, not on the evidence. Avogadro 2 computes its ESP from a charge model rather than from a second cube and recommends isovalues an order of magnitude above 0.001 a.u., so it fails R2 in substance and R4 outright. GaussView 6 does the mapping well but is commercial and has no scripting interface for the visualization side, failing R4 and R5 together.

1.3 Assessment

Program	R1 cube	R2 mapping	R3 scale	R4 script	R5 free	Outcome
PyMOL (open source)	yes	yes	yes	yes	yes	Selected — primary pipeline
VMD	yes	yes	yes	yes	(yes)	Selected — second pipeline
UCSF ChimeraX	yes	yes	yes	yes	(yes)	Viable, not built
Avogadro 2	yes	partly	no	no	yes	Fails R2, R3, R4
GaussView 6	yes	yes	partly	no	no	Fails R4 and R5

Table 3: (yes) marks a licence that is free of charge for academic use but carries restrictions: registration, academic-only terms, or a separate agreement for commercial use.

1.4 Why PyMOL and VMD

Three programs: PyMOL, VMD and ChimeraX satisfy all five requirements in substance. On the published criteria alone any of the three would have produced a defensible workflow, so the choice between them was not made by Table 3.

PyMOL was chosen first, and the honest reason is availability. It was already installed and in use from the introductory exercises of this course, so the time not spent installing and learning a second program went into the workflow instead. That is a practical argument rather than a technical one.

VMD was added as a second pipeline for two reasons. It is named explicitly in the project brief and two independent viewers reading the same cube files are a control on each other. Isosurface extraction, interpolation of the potential onto the surface vertices and the mapping of values to colours are three places where a plausible-looking but wrong picture can be produced and a single implementation cannot detect any of them.